

1.Connect your laptop and a friend's phone to the same Wi-Fi network, then use the ping command to check if both devices can communicate with each other.

Hint: On Windows, use 'ping [IP address]' in Command Prompt; on Android, try a ping app or use Termux.

Connect your laptop and a friend's phone to the same Wi-Fi network, then use the ping command to check if both devices can communicate with ip address.

```
Command Prompt
Microsoft Windows [Version 10.0.26200.9457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\DELL>ping 192.168.17.108

Pinging 192.168.17.108 with 32 bytes of data:
Reply from 192.168.17.133: Destination host unreachable.
Reply from 192.168.17.133: Destination host unreachable.
Reply from 192.168.17.133: Destination host unreachable.
Reply from 192.168.17.133: Destination host unreachable.

Ping statistics for 192.168.17.108:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

C:\Users\DELL>
```

2.Run the traceroute (tracert on Windows or traceroute on Mac/Linux) command to www.youtube.com from your device and note down the number of hops it takes to reach the destination.

```
Command Prompt
Microsoft Windows [Version 10.0.26200.9457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\DELL>tracert www.youtube.com

Tracing route to www.youtube.com [142.251.156.4]
over a maximum of 30 hops:

  1     6 ms    22 ms     5 ms    192.168.17.1 [192.168.17.1]
  2     6 ms     3 ms     3 ms    gpon.net [192.168.1.1]
  3    16 ms     8 ms     9 ms    10.130.192.1 [10.130.192.1]
  4     6 ms     5 ms     5 ms    10.111.13.5 [10.111.13.5]
  5      *      *      *      Request timed out.
  6    33 ms    32 ms    34 ms    142.251.156.4

Trace complete.
```

3.Disconnect one device from the Wi-Fi, then try pinging it again from your laptop and describe the error message you receive.

```
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C:\Users\DELL>ping 192.168.17.108

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Ping statistics for 192.168.17.108:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

4.Create a simple table listing the IP addresses of all devices connected to your home Wi-Fi (use 'arp -a' on Windows), and identify which device belongs to which person or gadget.

```
Command Prompt
Microsoft Windows [Version 10.0.26200.9457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\DELL>arp -a

Interface: 192.168.17.133 --- 0x11
    Internet Address      Physical Address      Type
    192.168.17.1          84-d8-1b-b0-97-32    dynamic
    192.168.17.255        ff-ff-ff-ff-ff-ff    static
    224.0.0.22            01-00-5e-00-00-16    static
    224.0.0.251           01-00-5e-00-00-fb    static
    224.0.0.252           01-00-5e-00-00-fc    static
    239.255.255.250       01-00-5e-7f-ff-fa    static
    255.255.255.255       ff-ff-ff-ff-ff-ff    static
```

5.Write a one-page report explaining step-by-step how you set up the network, tested connectivity using ping and traceroute, and what results or issues you observed.

Hint: Include screenshots or command outputs if possible.

```
Command Prompt
Microsoft Windows [Version 10.0.26200.9457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\DELL>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::183:2fb0:9291:2fb%17
    IPv4 Address. . . . . : 192.168.17.133
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.17.1

Ethernet adapter Bluetooth Network Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :
```

```
Command Prompt
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C:\Users\DELL>ping 192.168.17.108

Pinging 192.168.17.108 with 32 bytes of data:
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Reply from 192.168.17.133: Destination host unreachable.

Ping statistics for 192.168.17.108:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

```
Command Prompt
Microsoft Windows [Version 10.0.26200.9457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\DELL>tracert google.com

Tracing route to google.com [142.250.134.138]
over a maximum of 30 hops:

  1    6 ms    2 ms    3 ms  192.168.17.1 [192.168.17.1]
  2   11 ms    8 ms   62 ms  gpon.net [192.168.1.1]
  3    6 ms   12 ms   14 ms  10.130.192.1 [10.130.192.1]
  4    *    66 ms    4 ms  10.111.13.5 [10.111.13.5]
  5    *      *      *    Request timed out.
  6   42 ms   28 ms   28 ms  fx-in-f138.1e100.net [142.250.134.138]

Trace complete.
```

```
Command Prompt
Microsoft Windows [Version 10.0.26200.9457]
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C:\Users\DELL>arp -a

Interface: 192.168.17.133 --- 0x11
    Internet Address      Physical Address         Type
    192.168.17.1          84-d8-1b-b0-97-32       dynamic
    192.168.17.255        ff-ff-ff-ff-ff-ff       static
    224.0.0.22            01-00-5e-00-00-16       static
    224.0.0.251           01-00-5e-00-00-fb       static
    224.0.0.252           01-00-5e-00-00-fc       static
    239.255.255.250       01-00-5e-7f-ff-fa       static
    255.255.255.255       ff-ff-ff-ff-ff-ff       static
```